

Power BI Module end project- Fleet Performance and Delivery Efficiency Dashboard

Scenario: A logistics company wants to analyze its fleet performance in terms of on-time deliveries, fuel efficiency, and cost per mile

Dataset:

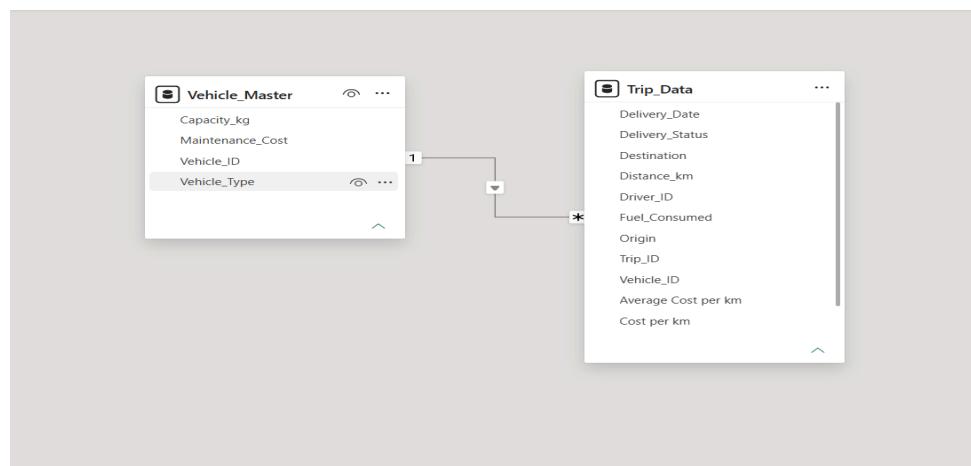
- Trip_data: Trip ID, Vehicle ID, Driver ID, Origin, Destination, Distance (km), Fuel Consumed (liters), Delivery Status (On-Time/Late), Delivery Date
- Vehicle Master: Vehicle ID, Vehicle Type, Capacity, Maintenance Cost.

Tasks: 1. Data Cleaning & Modeling:

- Fix missing fuel consumption values (use mean imputation).

A	B	C	D	E	F	G	H	I	J
Trip_ID	Vehicle_ID	Driver_ID	Origin	Destination	Distance_km	Fuel_Consumed	Fuel_Consumed_Update	Delivery_Status	Delivery_Date
T001	V04	D01	Delhi	Pune	1173	108.42	108.42	On-time	27-01-2023
T002	V06	D08	Mumbai	Bangalore	1727	161.33	161.33	On-time	21-02-2023
T003	V06	D08	Mumbai	Pune	1459	154.7	154.7	On-time	17-02-2023
T004	V04	D09	Hyderabad	Pune	382	26.6	26.6	On-time	18-02-2023
T005	V06	D08	Pune	Mumbai	398	33.2	33.2	On-time	15-02-2023
T006	V06	D07	Chennai	Mumbai	1275	85.04	85.04	Late	25-02-2023
T007	V07	D03	Chennai	Kolkata	752	58.08	58.08	On-time	19-01-2023
T008	V02	D10	Delhi	Pune	74	5.24	5.24	On-time	01-01-2023
T009	V02	D07	Delhi	Hyderabad	186	16.22	16.22	On-time	23-02-2023
T010	V02	D02	Bangalore	Hyderabad	1375	105.21	105.21	Late	02-02-2023
T011	V06	D03	Kolkata	Hyderabad	419	31.17	31.17	On-time	21-01-2023
T012	V06	D01	Kolkata	Delhi	751	51.77	51.77	On-time	15-02-2023
T013	V05	D04	Chennai	Hyderabad	1571	188.52	188.52	Late	02-02-2023
T014	V05	D05	Hyderabad	Bangalore	1524	104.51	104.51	On-time	16-02-2023
T015	V05	D06	Kolkata	Mumbai	1956	179.88	179.88	On-time	21-01-2023
T016	V05	D06	Bangalore	Mumbai	858	91.88	91.88	Late	18-01-2023
T017	V07	D07	Pune	Kolkata	1269	102.91	102.91	On-time	16-01-2023
T018	V07	D10	Pune	Delhi	1565	107.23	107.23	On-time	12-02-2023
T019	V07	D10	Hyderabad	Pune	1796	155.52	155.52	On-time	28-02-2023
T020	V04	D02	Mumbai	Bangalore	1640	148.87	148.87	On-time	06-01-2023
T021	V07	D07	Delhi	Mumbai	446	31.98	31.98	Late	11-01-2023
T022	V07	D09	Hyderabad	Delhi	140	11.44	11.44	On-time	07-02-2023
T023	V06	D04	Chennai	Delhi	1729	197.55	197.55	On-time	06-02-2023
T024	V04	D01	Kolkata	Mumbai	1376	96.63	96.63	On-time	14-02-2023
T025	V04	D02	Bangalore	Chennai	1721	120.65	120.65	Late	06-02-2023
T026	V05	D04	Chennai	Pune	1784	91.88	91.88	Late	17-01-2023
T027	V06	D07	Pune	Kolkata	1007	116.6	116.6	Late	17-02-2023
T028	V05	D06	Pune	Bangalore	1903	151.18	151.18	Late	20-02-2023
T029	V01	D06	Mumbai	Bangalore	1262	100.72	100.72	On-time	28-01-2023

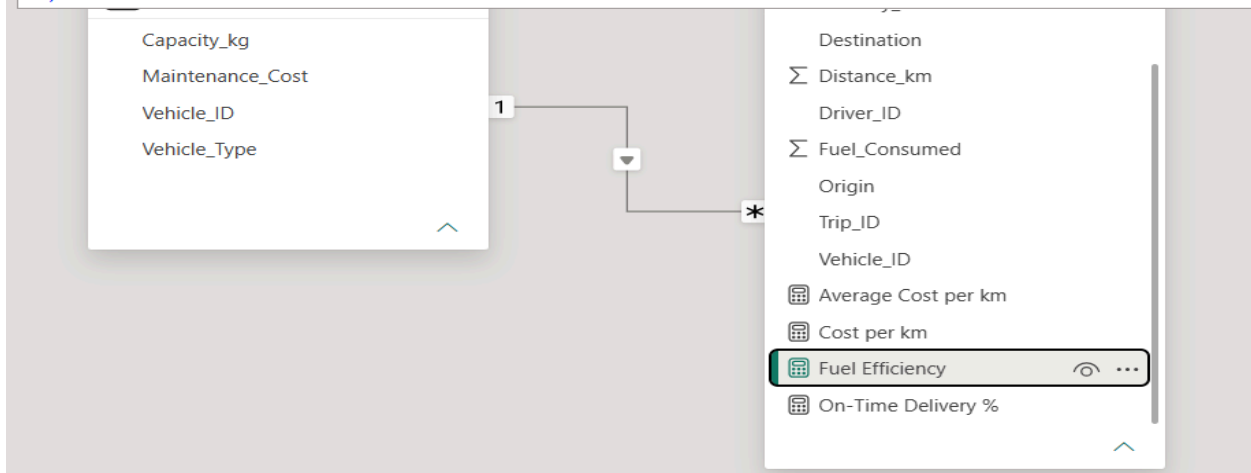
- Create a relationship using Vehicle_ID between Trip_Data and the Vehicle Master table.



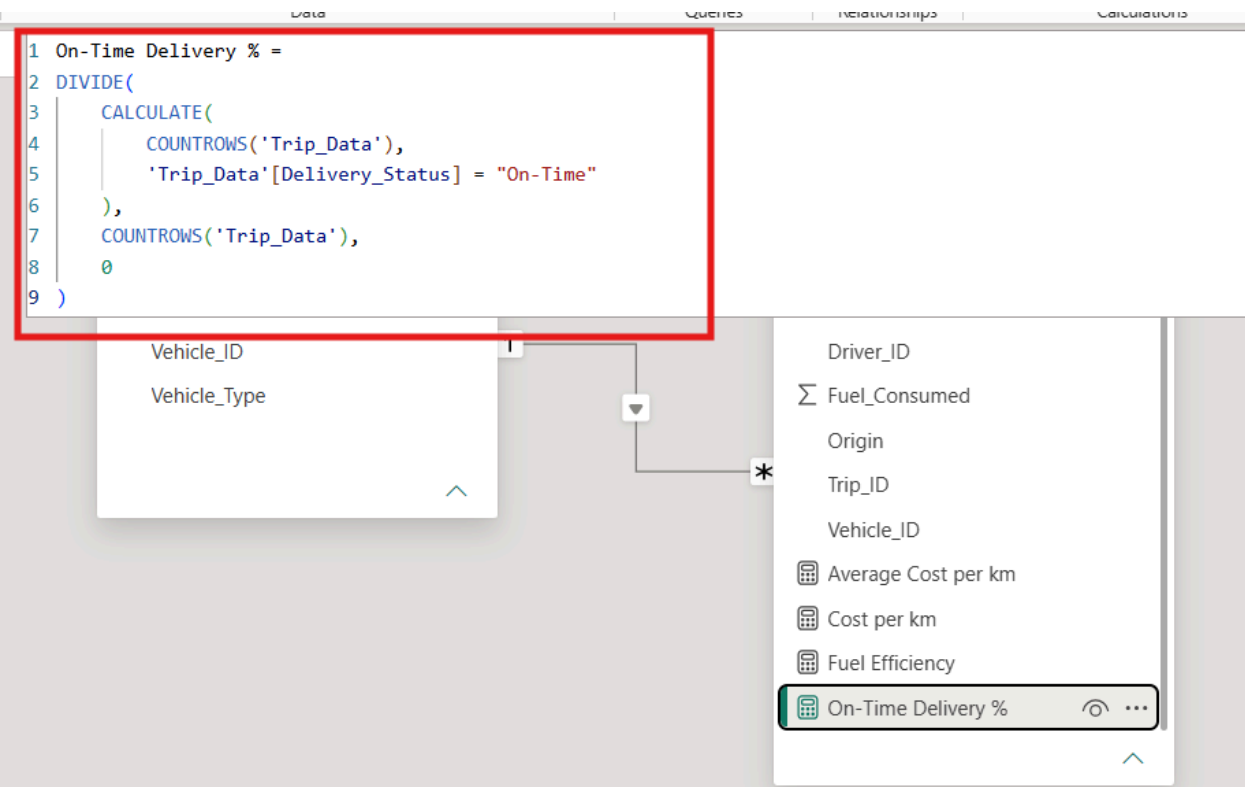
2. DAX Measures:

- **Fuel Efficiency** = Distance / Fuel Consumed

```
1 Fuel Efficiency =  
2 DIVIDE(  
3     SUM('Trip_Data'[Distance_km]),  
4     SUM('Trip_Data'[Fuel_Consumed]),  
5     0  
6 )
```



- **On-Time Delivery %** = On-Time Trips / Total Trips: To do this, calculate the number of "On-Time" entries in the table and the total entries in the Trip_Data table, and then divide both results.

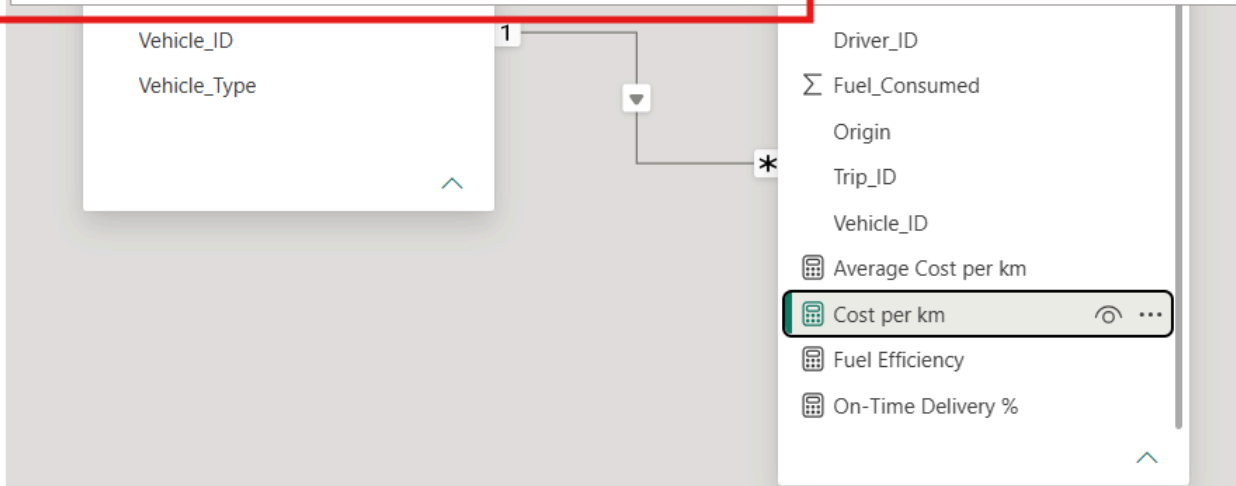


- **Cost per km** = (fuel cost + Maintenance Cost) / Distance Note fuel cost =100

```

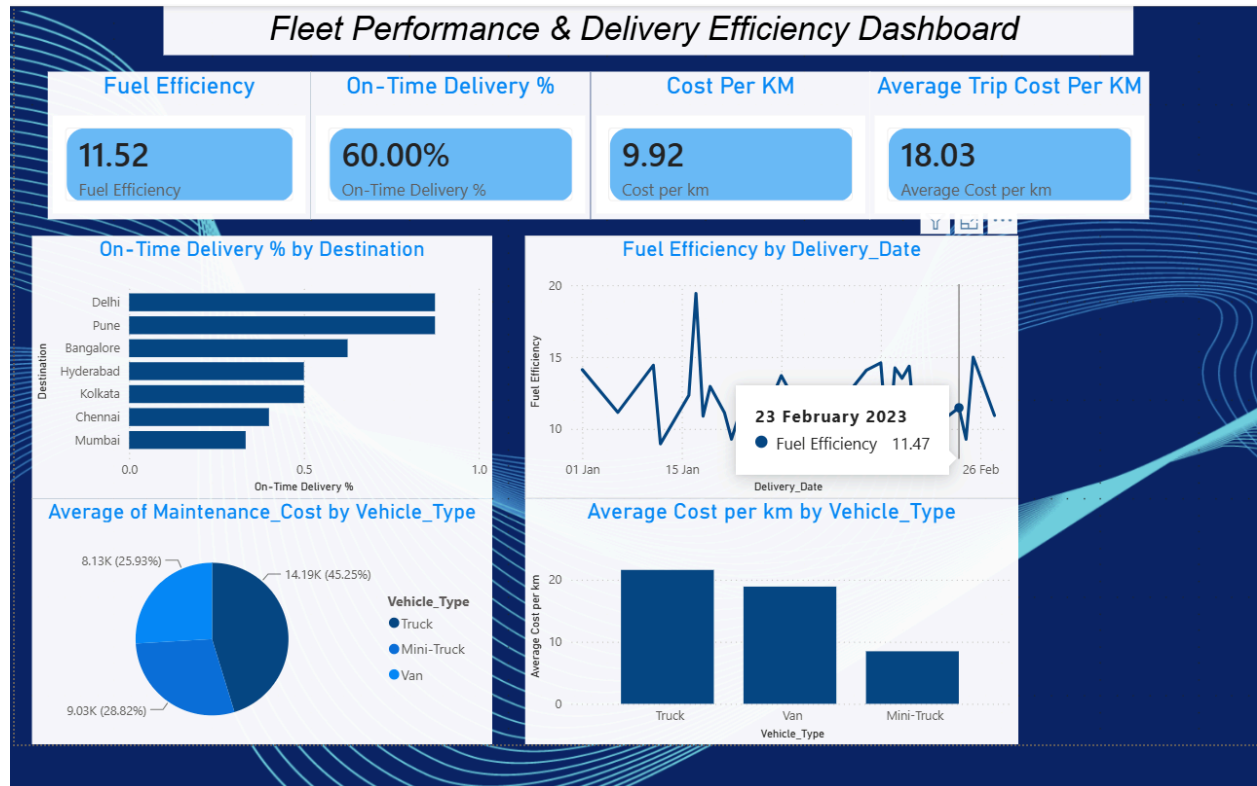
1 Cost per km =
2 DIVIDE(
3     SUMX(
4         'Trip_Data',
5         100 + RELATED('Vehicle_Master'[Maintenance_Cost])
6     ),
7     SUM('Trip_Data'[Distance_km]),
8     0
9 )

```



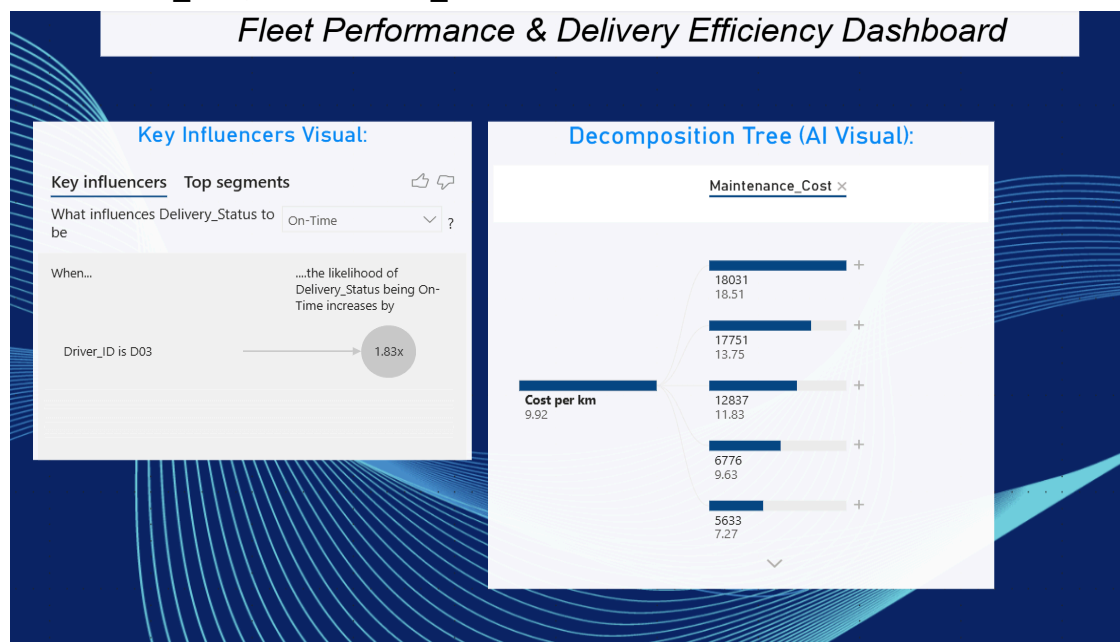
3. Visualization:

- Bar chart: On-Time Delivery % by Destination.
- Line chart: Fuel efficiency trend by delivery date.
- Cards visualization:
 - Avg. Delivery Time
 - Average cost per km
- Pie chart: vehicle type vs Average maintenance cost



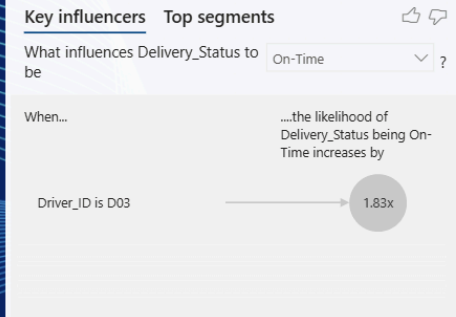
4. AI-Powered Visuals

- ❖ Q&A Visual: ➤ Add Q&A visual → Prompt: “Average Cost per km by vehicle type?”
- ❖ Key Influencers Visual: ➤ Add Delivery Status in the Analyze and explain by -distance in km, vehicle_type, Driver_ID
- ❖ Decomposition Tree (AI Visual): ➤ Analyze Cost per km → explained by Vehicle Type, Maintenance_cost, and Distance_km.



Fleet Performance & Delivery Efficiency Dashboard

Key Influencers Visual:

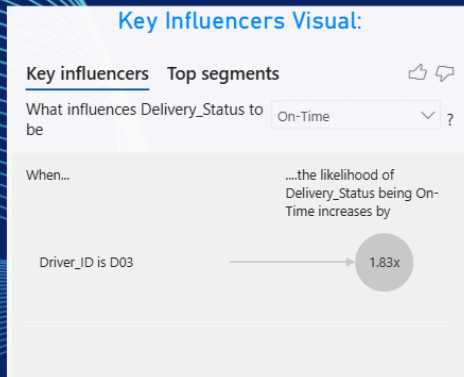


Decomposition Tree (AI Visual):



Fleet Performance & Delivery Efficiency Dashboard

Key Influencers Visual:



Decomposition Tree (AI Visual):

